regardless of their location or status within the data center, would receive the data at the same time.[821] In the Proposing Release, the Commission described latency neutralization protocols that exchanges could adopt, such as equal cable length protocols where cable lengths from network equipment to customer cabinets are harmonized for equal access. Any differential treatment of competing consolidators in the transmission of consolidated market data within a data center controlled by an exchange must be filed with the Commission as a proposed rule change pursuant to Section 19(b) of the Exchange Act and Rule 19b-4 thereunder and satisfy statutory requirements, such as not being designed to permit unfair discrimination[822] nor impose a burden on competition.[823]

(e) Comments on SRO Costs

One commenter, FINRA, stated that the ADF would be required to connect and provide data to all competing consolidators and self-aggregators.[824] FINRA said that it potentially could incur significant costs to establish and maintain this required connectivity, despite minimal fee revenue from data disseminated from the ADF given the low volume of regularly reported trades and lack of quoting participants.[825] FINRA stated in its comment letter that the ADF has a low

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821 See Proposing Release, 85 FR at 16771. For example, exchanges could adopt equal cable length protocols (i.e., where cross-connect cable lengths from network equipment to customer cabinets are harmonized for equal access) and ports of the same bandwidth (i.e., 1G, 10G, and 40G) to ensure that all of the exchange's data center connections provide market data simultaneously to market participants located within a data center.

822 Section 6(b)(5) of the Exchange Act, 15 U.S.C. 78f(b)(5).

823 Section 6(b)(8) of the Exchange Act, 15 U.S.C. 78f(b)(8).

824 See FINRA Letter at 4.

825 See id.

volume of over-the-counter ("OTC") trades and no quotes. Therefore, the connectivity options would not need to support much data capacity, which should limit the amount of costs incurred by FINRA to establish such connectivity. However, FINRA could seek to recoup any costs associated with establishing and maintaining connectivity to competing consolidators and self-aggregators by proposing connectivity fees pursuant to Section 19(b) of the Exchange Act.[826] Information about OTC quotations and trades in NMS stocks are an important component of SIP data today and will continue to be important in consolidated market data. This OTC information must be collected, consolidated, and disseminated in the decentralized consolidation model.

(f) Interpretation of Rule 603(a)

Currently, Rule 603(a) requires that SROs distribute NMS information on terms that are fair and reasonable and not unreasonably discriminatory.[827] The Commission stated in the Regulation NMS Adopting Release that Rule 603(a) would prohibit an SRO from making its core data available to vendors on a more timely basis than it makes such data available to the exclusive SIPs.[828] In particular, the Commission said that "independently distributed data could

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826 Connectivity to the data underlying consolidated market data would be a new data service because such service does not currently exist. The SROs will have to file with the Commission any proposed new fees for connectivity to their individual data that underlies consolidated market data pursuant to Section 19(b) of the Exchange Act and Rule 19b-4 thereunder, and any proposed connectivity fee must satisfy the statutory standards, including being fair and reasonable, being not unreasonably discriminatory, and reflecting an equitable allocation of reasonable fees. See Sections 6(b)(4) and 15A(b)(5) of the Exchange Act. See also infra note 1158 and accompanying text and Section III.E.2(c).

827 See Rule 603(a)(2) of Regulation NMS, 17 CFR 242.603(a)(2).

828 See Regulation NMS Adopting Release, supra note 7, at 37569. Specifically, the Commission stated, "Rule 603(a)(2) requires that any SRO, broker, or dealer that distributes market information must do so on terms that are not unreasonably discriminatory. These requirements prohibit, for example, a market from making its

not be made available on a more timely basis than core data is made available to a Network processor. Stated another way, adopted Rule 603(a) prohibits an SRO or broker-dealer from transmitting data to a vendor or user any sooner than it transmits data to a Network processor."[829] Today, latency differentials of any measurable amount are meaningful for certain market participants and their trading strategies. The Commission believes that Rule 603(a) prohibits an SRO from making NMS information available to any person on a more timely basis (i.e., by any time increment that could be measured by the SRO[830]) than it makes such data available to the existing exclusive SIPs and as amended, competing consolidators and self-aggregators. When it adopted Regulation NMS, the Commission did not provide a de minimis exception to the timeliness of data availability in Rule 603(a). If an SRO can measure the timeliness of data transmission in a specific increment, such increment generally should be utilized for determining whether such data has been transmitted on a more timely basis to persons other than an existing exclusive SIP, competing consolidators, or self-aggregators.

'core data' (i.e., data that it is required to provide to a Network processor) available to vendors on a more timely basis than it makes available the core data to a Network processor." Id.

829 Id. at 37567.

830 For example, latency is currently measured for SIP data in microsecond increments. See CTA Key Operating Metrics of Tape A&B U.S. Equities Securities Information Processor (CTA SIP), available at https://www.ctaplan.com/publicdocs/ctaplan/CTAPLAN_Processor_Metrics_3Q2020.pdf (last accessed Nov. 27, 2020).

10. Calculation of the National Best Bid and National Best Offer under Rule 600(b)(50)

(a) Proposal

The Commission proposed to amend the definition of national best bid and national best offer to reflect that competing consolidators and self-aggregators, rather than the exclusive SIPs, would be calculating the NBBO in the proposed decentralized consolidation model. In addition, to accommodate this proposed decentralized consolidation model, the Commission proposed to bifurcate the NBBO definition between NMS stocks and other NMS securities (i.e., listed options) to reflect that the proposed decentralized consolidation would apply only with regard to NMS stocks, and therefore the plan processor for options would continue to be responsible for calculating and disseminating the NBBO in listed options. The proposed changes to the definition of NBBO would not impact the manner in which the NBBO is calculated for NMS stocks.[831]

(b) Comments on Complexity and Confusion Resulting from Multiple NBBOs

The Commission received multiple comments with respect to the proposed definition of NBBO, with commenters expressing concerns about multiple NBBOs and the impact on market participants and investors. Commenters stated that confusion could result from multiple NBBOs being available to market participants.[832] One commenter stated that retail investors do not have

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831 The NBBO will reflect the new round lot sizes. See supra Section II.D.2.

832 See TD Ameritrade Letter at 12 ("In the present-day structure, SIPs provide a 'gold copy' standard of the NBBO which can be used for comparison when meeting the regulatory requirements of best execution and the Vendor Display Rule…The industry may have as many NBBO quotes as there are [competing consolidators] and self-aggregators."); Healthy Markets Letter I at 4 ("We have significant concerns that by creating multiple,

experience with multiple NBBOs.[833] The commenter further noted that multiple NBBOs would make broker-dealers' Rule 605 reports difficult to compare because broker-dealers could be using different NBBOs.[834] The commenter questioned whether a broker-dealer would be responsible for execution quality and compliance with the Vendor Display Rule if the competing consolidator that the broker-dealer is using miscalculates the NBBO.[835]

Other commenters noted that a lack of a single NBBO as calculated by the existing SIPs could impact the market in a variety of ways, such as a lack of a single benchmark to determine trade quality; curtailing the Commission's surveillance efforts; potentially creating a multi-tiered system where some traders can access faster NBBOs;[836] and adding significant complexity,

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alternative SIPs, that market participants may cherry-pick the data feed that costs them the least or makes them look the best."); Nasdaq Letter IV at 12; Fidelity Letter at 10–11.

833 See TD Ameritrade Letter at 12–13 ("Furthermore, even if proprietary feeds disseminate a different quote than the NBBO, such quote has never been permitted for purposes of use in the Vendor Display Rule, meaning the retail investor has not previously experienced a multiple NBBO environment."). See also Fidelity Letter at 10–11 ("Under the Proposal, retail customers placing orders in the same security at two different brokerage firms may receive two different prices depending upon which competing consolidator each firm uses."). This scenario could happen today. The Commission does not believe the decentralized consolidation model will introduce confusion for retail investors. Retail investors will continue to be able to see the NBBO provided by their broker-dealer, whether it is calculated by their broker-dealer or a competing consolidator.

834 See TD Ameritrade Letter at 13.

835 See id. The commenter further questioned whether regulators would grant relief when reviewing execution quality on an order by order basis, when different competing consolidators are calculating NBBOs differently.

836 See Nasdaq Letter IV at 7 ("Shedding the single NBBO – which has long been investors' 'North Star' for price discovery – in favor of multiple NBBOs would further complicate market structure, confuse investors as to whether they are actually seeing the best price, and hinder market surveillance and enforcement efforts.").

confusion, and uncertainty to best execution analysis.[837] One commenter said that it is unlikely the NBBOs produced by competing consolidators and their associated timestamps would ever be synchronized and that the NBBOs calculated by competing consolidators and self-aggregators will never align. The commenter said that the different versions of market data would impact broker-dealer compliance with market conduct rules.[838]

Another commenter noted that multiple NBBOs would make comparison of broker-dealer performance difficult, as broker-dealers may decide which NBBO to use depending on which NBBO costs the least or makes their execution quality look better.[839] The commenter acknowledged that while there is no clear NBBO today, the exclusive SIPs do calculate and disseminate a NBBO.[840] Commenters questioned how execution quality would be judged if each competing consolidator provided a different NBBO.[841] Another commenter believed that multiple NBBOs will impact liquidity.[842] One commenter said that latency arbitrage would be required to "keep markets in line," estimating an increase of almost $3 billion in latency

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837 See FINRA Letter at 6.

838 See id. at 3–4.

839 See Healthy Markets Letter I at 4.

840 See id.

841 See Schwab Letter at 6; NYSE Letter II at 24; WFE Letter at 1.

842 See Data Boiler Letter I at 3. The commenter did not describe how it thought multiple NBBOs would impact liquidity. As discussed, the current market operates with multiple NBBOs. The Commission believes that the NBBOs that will be calculated in the decentralized consolidation model better reflect current market conditions due to the improvements in latencies and therefore, will be more prompt, accurate, and reliable. The enhanced and faster consolidated market data resulting from the proposal should allow market participants access to more up-to-date market data (including NBBO data) than provided by the SIP, permitting them to more readily find liquidity than before.

arbitrage profits as a result of the multiple NBBOs introduced by the proposal.[843] This commenter said that multiple NBBOs would also reduce investor confidence that trades will be executed at the best price[844] and believed that multiple NBBOs would negatively impact market quality, complicate best execution compliance by broker-dealers, and overall hurt investor confidence in the markets.[845] Another commenter requested clarity that a single competing consolidator could produce multiple NBBOs at different locations at any given point of time, since some competing consolidators may co-locate at more than one exchange data centers.[846]

In the decentralized consolidation model, competing consolidators will replace the exclusive SIPs in generating the NBBO as defined in Rule 600. Therefore, there will be NBBOs generated by each competing consolidator, as well as self-aggregators, based upon the data content received from the SROs pursuant to Rule 603(b). While commenters expressed concerns about the loss of an NBBO generated by a single plan processor—the exclusive SIPs—the Commission believes that market participants will adjust to not having an NBBO generated by a single plan processor. "Multiple NBBOs" already exist today.[847] Some market participants obtain the NBBO from the exclusive SIPs, and some market participants generate their own

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843 Nasdaq Letter III at 8. Many firms that aggregate proprietary feeds today likely have different aggregation times and are likely already exposed to latency arbitrage. It is unclear why the use of latency arbitrage would increase as a result of the proposal, especially since the existence of multiple NBBOs would not represent a change from current market practices.

844 See Nasdaq Letter IV at 11.

845 See id. at 3.

846 See McKay Letter at 10–11 ("…many market participants are colocated at one or more of the major exchange datacenters in New Jersey and receive their market data at such colocated points of presence. If a competing consolidator seeks to provide market participants with the fastest and most efficient consolidated market data possible, the result would be a slightly different NBBO at each exchange datacenter.").

847 See also infra Section V.B.2(f)

NBBO by aggregating multiple proprietary data feeds. Some market participants that generate their own NBBO use third party aggregation software or services to do so. The NBBO seen by market participants depends on the systems used to generate the NBBO and the systems used by the market participants to receive and view it. Market participants are therefore already accustomed to a market environment in which there are "multiple NBBOs" generated by different parties, at different speeds, in different locations. They are accustomed to performing best execution analysis and providing execution quality statistics in the current environment. Therefore, the Commission does not believe that the generation of NBBOs by multiple competing consolidators will add complexity or confusion to the markets.

Moreover, neither the amended definition of NBBO, nor the decentralized consolidation model more generally, mandates the consumption of multiple NBBOs from multiple competing consolidators. So, while different competing consolidators may calculate and disseminate unique NBBOs or single competing consolidators may calculate different NBBOs at separate data center locations under the decentralized consolidation model, the Commission does not believe that this should cause confusion for market participants, who already consider market data from multiple sources. Finally, the amended definition of NBBO does not change the methodology by which the NBBO for a particular NMS stock is calculated, which will help to ensure consistency in the NBBO calculation. The Commission is therefore adopting the amendments to the definition of NBBO as proposed.

(c) Comments on Impact of Multiple NBBOs on Best Execution Obligations

With respect to best execution, commenters stated that the Commission did not sufficiently discuss the best execution implications of multiple NBBOs.[848] Commenters stated that eliminating the single NBBO, and requiring broker-dealers to choose among multiple NBBOs, would complicate and increase the cost of compliance with best execution obligations.[849] However, another commenter stated that it "does not believe the existence of multiple NBBOs will create problems for market participants or the market as a whole at a level that would warrant not moving forward with the decentralized, competitive model…."[850]

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848 See, e.g., Nasdaq Letter IV at 3, 12–13; Clearpool Letter at 8 ("[W]hile the proposal states that the existence of multiple NBBOs does not impact a broker's best execution obligations, we believe that "fragmenting" the NBBO could lead to several problems around such obligations, which would need to be addressed and clarified by the Commission prior to implementation."); Healthy Markets Letter I at 4–5 ("Could a market participant seek to deliberately subscribe to an inferior data feed? … We urge the Commission to establish objective standards, including policies, procedures, and documentation requirements for brokers regarding their selection and usage of competing data feeds."); FINRA Letter at 5 ("[T]he Commission may also want to consider providing guidance on whether a broker-dealer would, or should, be evaluated—by the Commission, FINRA, or others—on its decision of which competing consolidator(s) to receive consolidated market data from, what factors a broker-dealer should consider in evaluating its choice of a competing consolidator(s) (both initially and on an ongoing basis), and how a broker-dealer's choice of a competing consolidator(s) might affect the broker-dealer's best execution obligations.").

849 See Nasdaq Letter IV at 3, 42, 49; Joint CRO Letter at 2; STANY Letter II at 6; TD Ameritrade Letter at 12–13.

850 Clearpool Letter at 8; see also Capital Group Letter at 4 ("We also support the proposed commercial competing consolidators. Different users have different needs with regards to data complexity (e.g., what data elements are included or excluded), latency and price. We do not see this as the creation of 'multiple truths' or 'multiple markets.' Because the data will also be available for post trade analysis, market participants will be able to analyze outcomes using the difference between the originating venue timestamps and the consolidator timestamps to determine if best execution obligations were fulfilled and if the speed they chose to pay for meets their needs.").

Additionally, one commenter stated that expanding core data generally would facilitate best execution.[851]

As described above, "multiple NBBOs" already exist today. The amended definition of NBBO will not significantly alter this state of affairs; rather, it adjusts the definition of NBBO to accommodate the decentralized consolidation model to reflect that NBBOs will be calculated and disseminated by competing consolidators and calculated by self-aggregators. As is the case today, broker-dealers must act reasonably to obtain pricing information in carrying out their duty to seek to obtain the most favorable terms reasonably available under the circumstances for the execution of customer orders.[852]

(d) Comments on Impact of Multiple NBBOs on Surveillance and Enforcement

While some commenters expressed concern about the proposed decentralized consolidation model's impact on regulation,[853] several commenters more specifically opined that the multiple NBBOs resulting from the proposed model would undermine exchange surveillance and enforcement,[854] and raised questions about how market participants should consider multiple NBBOs in trade-through prevention and locked and crossed markets.[855] One commenter said that the proposal did not fully consider the impact of multiple NBBOs on surveillance and

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851 See Better Markets Letter at 2.

852 See supra Section I.E.

853 See supra Section III.B.8.

854 See Joint CRO Letter at 2, 3; Nasdaq Letter IV at 12; NYSE Letter II at 24; see also TechNet Letter II at 2 (stating that multiple NBBOs would undermine effective regulation and investor confidence).

855 See Joint CRO Letter at 2, 3, 4; Nasdaq Letter IV at 3, 12, 36, 49.